

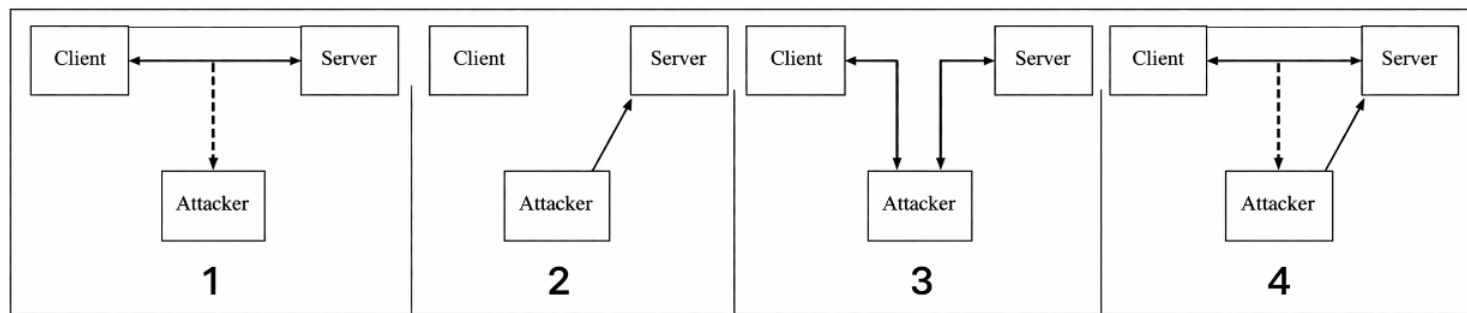


**CCY2001 Introduction to Cybersecurity (30 Points)**

Student ID: \_\_\_\_\_ Student Name: \_\_\_\_\_ Class: \_\_\_\_\_

**Question One: Multiple Choice Questions (10 points)**

The figure below shows four different attack scenarios labeled 1, 2, 3, and 4. Select the correct description of the attack type shown.



- Passive Attack, Masquerade, Replay, Denial-of-Service
  - Masquerade, Passive Attack, Denial-of-Service, Replay
  - Replay, Masquerade, Passive Attack, Denial-of-Service
  - Passive Attack, Replay, Masquerade, Denial-of-Service
- 2.** Which of the following statements correctly distinguishes the basic CIA triad from the extended CIAAA model in cybersecurity?
- CIA stands for Confidentiality, Integrity, Availability; CIAAA adds Authentication, Accountability, and Auditability.
  - CIA ensures data is encrypted; CIAAA only ensures physical security of servers.
  - CIAAA is a replacement of CIA; it ignores Integrity and Availability.
  - CIA focuses on network attacks; CIAAA focuses on social engineering attacks.
- 3.** What character sequence is often used in SQL injection attacks to bypass authentication?
- ' OR '1'='1
  - ' SELECT \* FROM products; –
  - '; DROP TABLE users; –
  - ' AND '2'=2
- 4.** Cross-site scripting (XSS) is a client-side code injection attack. Which of the following statements about XSS is correct?
- XSS attacks execute malicious code on the server, not the users browser.
  - XSS can allow attackers to steal a users cookies and impersonate them online.
  - XSS only affects network routers and firewalls.
  - Client-side code in XSS is typically written in SQL.
- 5.** A friend sends an electronic Hallmark greeting card (e-card) to your work email. You need to click on the attachment to see the card. What should you do? List major risks associated with this scenario.
- Click the attachment immediately; it is safe since it comes from a friend.
  - Verify the senders identity and scan the attachment with antivirus software before opening it.
  - Ignore the email entirely; there are no risks in opening unknown attachments.
  - Forward the attachment to colleagues to share the card quickly.

**Scenario:** A large financial organization discovers unusual activity in its network logs. Over the past week, some employees reported receiving suspicious emails containing attachments. Shortly afterward, a critical database became inaccessible for several hours, causing disruption to the organization’s operations. Fortunately, no sensitive customer data was leaked, but internal investigations show that an unauthorized user attempted to modify system files.

- 1: Which of the following represents a **Threat** in the scenario?
  - (a) The unauthorized user attempting to modify system files
  - (b) The suspicious emails received by employees
  - (c) The database becoming inaccessible
  - (d) All of the above
- 2: Which of the following is considered an **Attack**?
  - (a) Receiving a suspicious email
  - (b) Attempted modification of system files by an unauthorized user
  - (c) Potential disruption to operations
  - (d) Reporting the unusual activity
- 3: What is the key difference between a **Threat** and an **Attack**?
  - (a) A Threat is the actual malicious activity, whereas an Attack is only a potential risk
  - (b) A Threat is a potential risk to assets or operations, whereas an Attack is the actual malicious activity
  - (c) Threats always result in data loss, while Attacks never do
  - (d) Threats are external, and Attacks are always internal
- 4: Which of the following is a **major risk** if employees click on attachments from suspicious emails?
  - (a) Infection with malware or ransomware
  - (b) Unauthorized access to sensitive files
  - (c) Disruption of business operations
  - (d) All of the above
- 5: What is the best immediate action for employees who receive suspicious emails in this scenario?
  - (a) Open the attachment to see what it contains
  - (b) Forward the email to colleagues
  - (c) Report the email to IT/security team and delete it
  - (d) Ignore the email and continue work

**Question Two: Essay Question (20 points)**  
**Alphabet to Numerical Code Mapping**

| Letter | a | b | c | d | e | f | g | h | i | j | k  | l  | m  |
|--------|---|---|---|---|---|---|---|---|---|---|----|----|----|
| Code   | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |

| Letter | n  | o  | p  | q  | r  | s  | t  | u  | v  | w  | x  | y  | z  |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Code   | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

- 1- Encrypt the word **cryptographic** using the Vigenère cipher with the key **grp**.
- 2- Using the Row Transposition Cipher with key (4 3 1 2 5 6 7), encrypt the following text: **Secure the network before midnight**.
- 3- Suppose that the plaintext **back** is encrypted to the ciphertext **Iguq** using the Hill cipher; find the encryption key.
- 4- Using Playfair cipher with a key (**algorithm**), construct its matrix then encrypt the following sentence: **we are discovered save yourself**.
- 5- Decrypt the ciphertext "**owhl**" using the function  $D$  defined by  $D(C) = D_2(D_1(C))$ , where:
  - $D_1$  is the decryption of the Vigenère cipher with key (5, 10, 17).
  - $D_2$  is the decryption of the substitution cipher  $E_2$ .

**Substitution Table for  $E_2$**

| $E_2$  | a | b | c | d | e | f | g | h | i | j | k | l | m |
|--------|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher | x | n | y | a | h | p | o | g | z | q | w | b | t |

| $E_2$  | n | o | p | q | r | s | t | u | v | w | x | y | z |
|--------|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher | s | f | l | r | c | v | m | u | e | k | j | d | i |